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Пензенский государственный университет
Кафедра «Вычислительная техника»

Отчет

по лабораторной работе №3
по дисциплине «Программирование на языке Java»
на тему: «Обработка исключительных ситуаций»
Вариант 6

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Цель работы

Изучить механизм обработки исключительных ситуаций.

Лабораторное задание

Модифицировать приложение из предыдущей лабораторной работы, реализовав проверку вводимых данных с использованием механизма исключений. Необходимо создать свой класс, унаследованный от класса `Exception`, и генерировать исключение, если возникает попытка создать экземпляр класса `RecIntegral` со значениями, не являющимися числами в диапазоне от 0,000001 до 1000000. В качестве обработки исключения необходимо выводить диалог, содержащий предупреждение о некорректности введенных данных. Оформление лабораторной работы должно быть выполнено в соответствии с требованиями, приведенными в Приложении 2.

Ход работы:

1. Реализовали класс для исключительных ситуаций `InvalidValueException`, унаследованный от класса `Exception`.

```
*/  
public class InvalidValueException extends Exception {  
    public InvalidValueException(String description){  
        JOptionPane.showMessageDialog(null, description, "ERROR!!!", JOptionPane.ERROR_MESSAGE);  
    }  
}
```

Рисунок 1 - класс `InvalidValueException`

2. Добавили в конструктор класса `RecIntegral` блоки `try-catch` для проверки исключительных ситуаций при создании экземпляра класса.

```

public RecIntegral(double botLimit, double topLimit, double step) throws InvalidValueException {
    if (botLimit < 0.000001 || botLimit > 1000000)
        throw new InvalidValueException("The value of the bottom limit of the integral must be between 0.000001 and 1000000");

    if (topLimit < 0.000001 || topLimit > 1000000)
        throw new InvalidValueException("The value of the top limit of the integral must be between 0.000001 and 1000000");

    if (step < 0.000001 || step > 1000000)
        throw new InvalidValueException("The value of the step of the integral must be between 0.000001 and 1000000");

    if (botLimit >= topLimit)
        throw new InvalidValueException("The value of the bottom limit must not be greater than the top limit!!!");

    this.botLimit = botLimit;
    this.topLimit = topLimit;
    this.step = step;
    this.result = 0.0;
}

```

Рисунок 2 - конструктор класса RecIntegral

Результат работы программы:

Current function is $\sqrt{x}$

Enter the bottom limit: 0.000001

Enter the top limit: 3

Enter the step: 0.001

Bottom limit	Top limit	Step	Result
0.000001	3	0.001	

ERROR!!!

The value of the bottom limit of the integral must be between 0.000001 and 1000000!!!

OK

Рисунок 3 - Ввод некорректного значения нижней границы интеграла

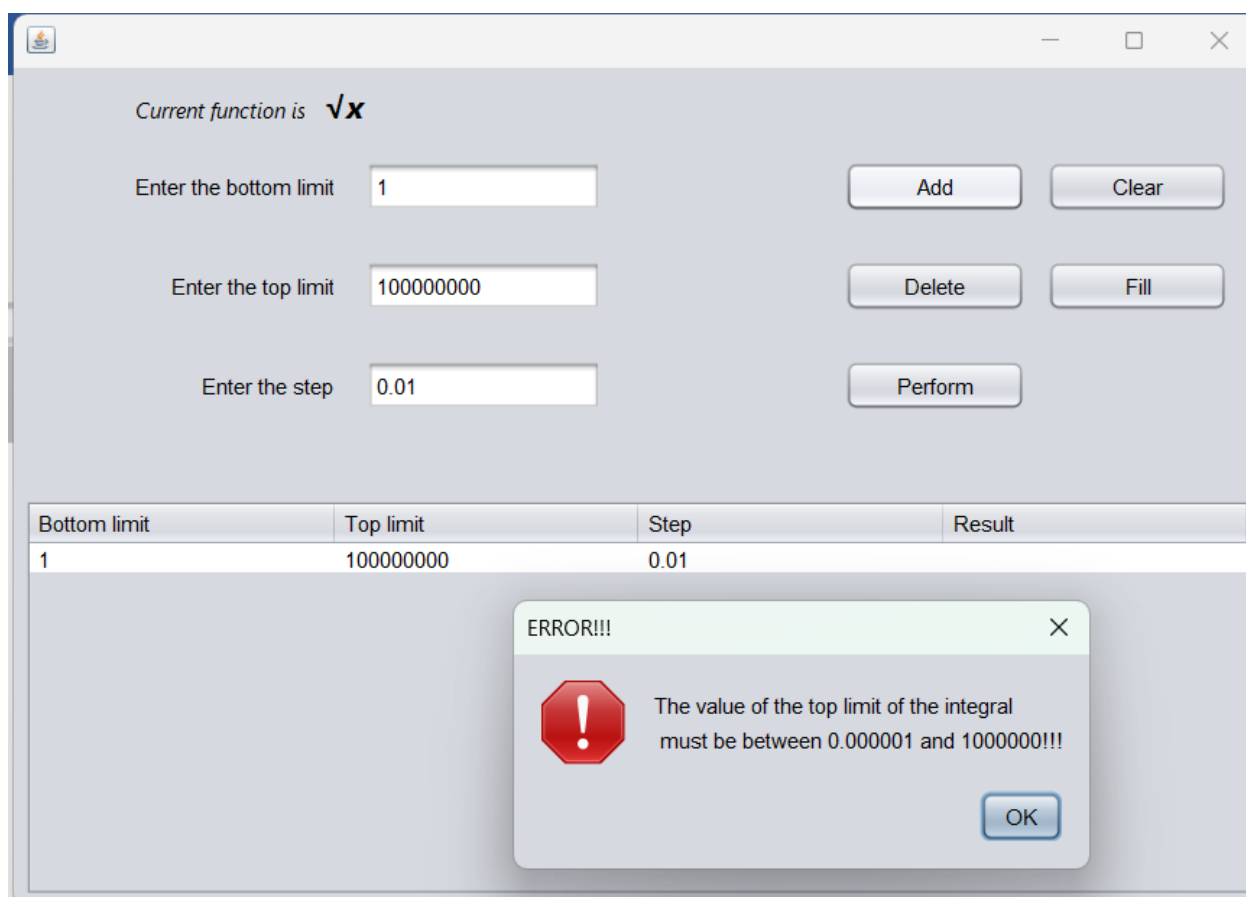


Рисунок 4 - Ввод некорректного значения верхней границы интеграла

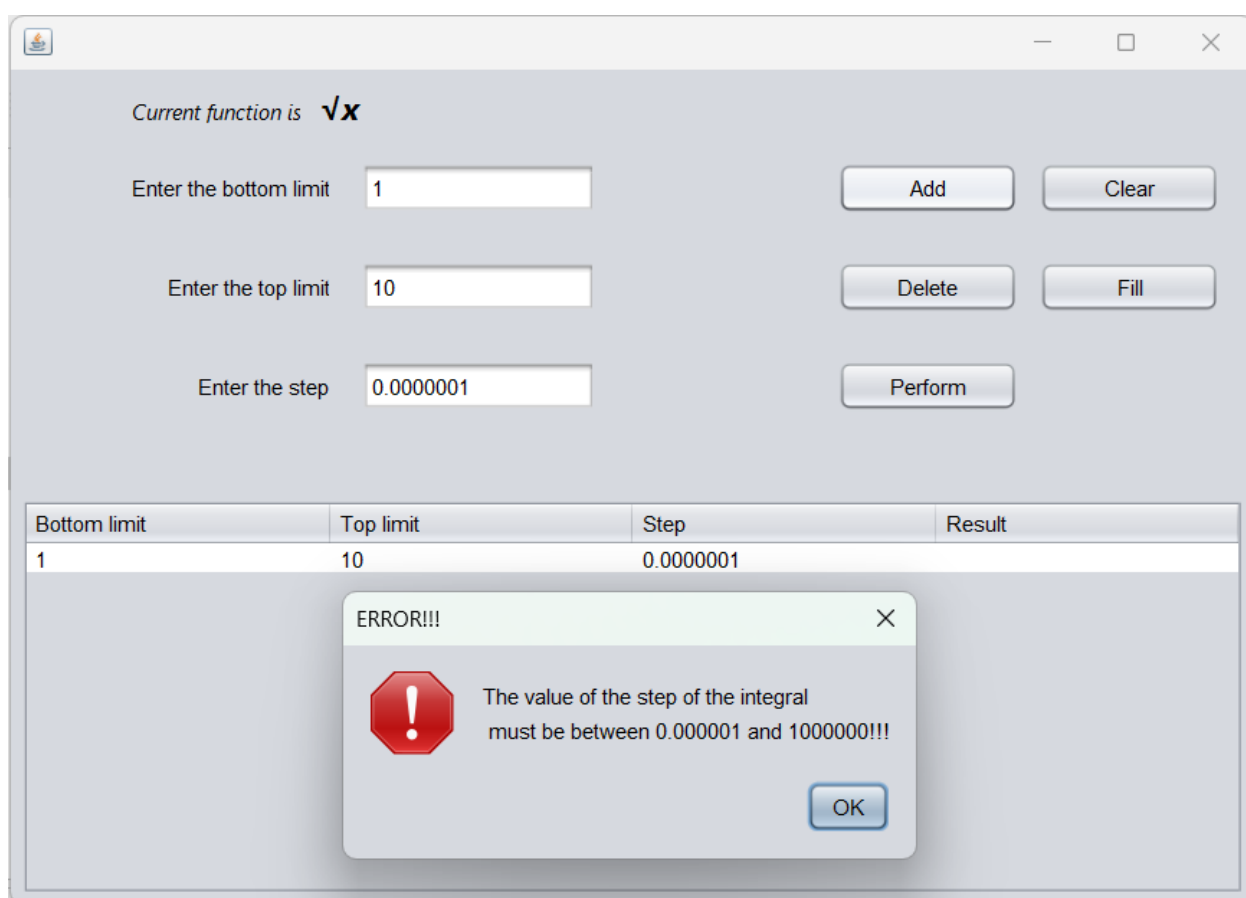


Рисунок 5 - Ввод некорректного значения шага

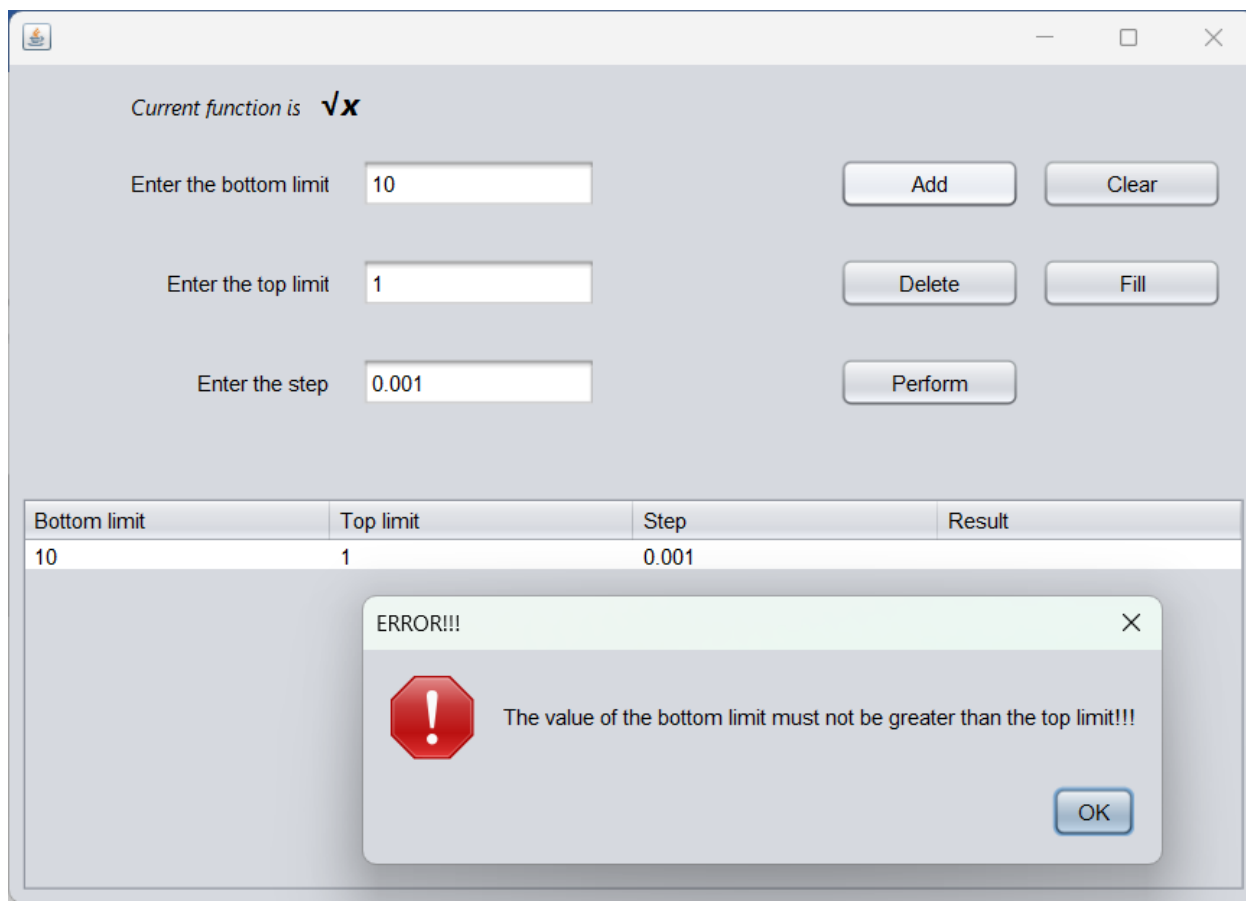


Рисунок 6 - Ввод значения, где значения нижней границы больше значения верхней границы

Вывод

При выполнении данной работы мы изучили механизм обработки исключительных ситуаций, а также научились вызывать и отлавливать исключения на практике.

Листинг

Файл ThirdTask.java

$$/*$$

* Click `nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt` to change this license

* Click `nbfs://nbhost/SystemFileSystem/Templates/GuiForms/JFrame.java` to edit this template

$$*/$$

```
package ru.psu.examplefirst;
```

```
import java.util.LinkedList;
```

```
import javax.swing.JOptionPane;
```

```
import javax.swing.table.DefaultTableModel;
```

/***

*

* @author Asus

*/

```
public class ThirdTask extends javax.swing.JFrame {
```

/**

* Creates new form FirstExample

 $\ast/$

```
LinkedList<RecIntegral> functionList;
```

```
public ThirdTask() {  
    initComponents();  
    functionList = new LinkedList<>();  
}
```

```
boolean isClear = true;
```

```
/**
```

```
 * This method is called from within the constructor to initialize the form.
```

```
 * WARNING: Do NOT modify this code. The content of this method is always
```

```
 * regenerated by the Form Editor.
```

```
 */
```

```
@SuppressWarnings("unchecked")
```

```
// <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents
```

```
private void initComponents() {
```

```
    jPanel = new javax.swing.JPanel();
```

```
    jScrollPane1 = new javax.swing.JScrollPane();
```

```
    jTable = new javax.swing.JTable();
```

```
txtBot = new javax.swing.JTextField();

txtTop = new javax.swing.JTextField();

txtStep = new javax.swing.JTextField();

labelBot = new javax.swing.JLabel();

labelTop = new javax.swing.JLabel();

labelStep = new javax.swing.JLabel();

btnAdd = new javax.swing.JButton();

btnDelete = new javax.swing.JButton();

btnPerform = new javax.swing.JButton();

labelFunction = new javax.swing.JLabel();

labelF = new javax.swing.JLabel();

btnClear = new javax.swing.JButton();

btnFill = new javax.swing.JButton();
```

```
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
```

```
jTable.setModel(new javax.swing.table.DefaultTableModel(
    new Object [][] {

    },
    new String [] {
        "Bottom limit", "Top limit", "Step", "Result"
    }
));
```



```

    }
) {

    boolean[] canEdit = new boolean [] {

        true, true, true, false

    };

    public boolean isCellEditable(int rowIndex, int columnIndex) {

        return canEdit [columnIndex];

    }

});

jScrollPane1.setViewportViewView(jTable);

labelBot.setText("Enter the bottom limit");

labelTop.setText("Enter the top limit");

labelStep.setText("Enter the step");

btnAdd.setText("Add");

btnAdd.addActionListener(new java.awt.event.ActionListener() {

    public void actionPerformed(java.awt.event.ActionEvent evt) {

        btnAddActionPerformed(evt);

    }
}

```

```
});
```

```
btnDelete.setText("Delete");
```

```
btnDelete.addActionListener(new java.awt.event.ActionListener() {
```

```
    public void actionPerformed(java.awt.event.ActionEvent evt) {
```

```
        btnDeleteActionPerformed(evt);
```

```
    }
```

```
});
```

```
btnPerform.setText("Perform");
```

```
btnPerform.addActionListener(new java.awt.event.ActionListener() {
```

```
    public void actionPerformed(java.awt.event.ActionEvent evt) {
```

```
        btnPerformActionPerformed(evt);
```

```
    }
```

```
});
```

```
labelFunction.setFont(new java.awt.Font("Segoe UI", 2, 12)); // NOI18N
```

```
labelFunction.setText("Current function is");
```

```
labelF.setFont(new java.awt.Font("Segoe UI", 3, 18)); // NOI18N
```

```
labelF.setText("  $\sqrt{x}$ ");
```

```
btnClear.setText("Clear");
```

```
btnClear.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnClearActionPerformed(evt);  
    }  
});
```

```
btnFill.setText("Fill");  
btnFill.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnFillActionPerformed(evt);  
    }  
});
```

```
javax.swing.GroupLayout jPanelLayout = new  
javax.swing.GroupLayout(jPanel);  
jPanel.setLayout(jPanelLayout);  
jPanelLayout.setHorizontalGroup(  
jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)  
    .addGroup(jPanelLayout.createSequentialGroup()  
        .addGap(10, 10, 10)  
        .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 600, true)  
    )  
);
```

```
.addContainerGap())

.addGroup(jPanelLayout.createSequentialGroup())

.addGap(67, 67, 67)

.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

.addGroup(jPanelLayout.createSequentialGroup())

.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

.addComponent(labelStep,
javax.swing.GroupLayout.Alignment.TRAILING)

.addComponent(labelBot,
javax.swing.GroupLayout.Alignment.TRAILING)

.addComponent(labelTop,
javax.swing.GroupLayout.Alignment.TRAILING))

.addGap(18, 18, 18)

.addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

.addComponent(txtBot,
javax.swing.GroupLayout.PREFERRED_SIZE, 130,
javax.swing.GroupLayout.PREFERRED_SIZE)

.addComponent(txtTop,
javax.swing.GroupLayout.PREFERRED_SIZE, 130,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```

        .addComponent(txtStep,
javax.swing.GroupLayout.PREFERRED_SIZE, 130,
javax.swing.GroupLayout.PREFERRED_SIZE)))

        .addGroup(jPanelLayout.createSequentialGroup()

        .addComponent(labelFunction)


        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)

        .addComponent(labelF,
javax.swing.GroupLayout.PREFERRED_SIZE, 37,
javax.swing.GroupLayout.PREFERRED_SIZE)))

        .addGap(135, 135, 135)


        .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignmen
t.LEADING, false)

        .addComponent(btnAdd, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

        .addComponent(btnDelete,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

        .addComponent(btnPerform,
javax.swing.GroupLayout.DEFAULT_SIZE, 100, Short.MAX_VALUE))

        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)


        .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignmen
t.LEADING, false)

```

```

        .addComponent(btnClear,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

        .addComponent(btnFill, javax.swing.GroupLayout.DEFAULT_SIZE,
100, Short.MAX_VALUE))

        .addContainerGap(20, Short.MAX_VALUE))

    );

    jPanelLayout.setVerticalGroup(

jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanelLayout.createSequentialGroup()

            .addGap(8, 8, 8)

        .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)

            .addComponent(labelFunction)

            .addComponent(labelF))

            .addGap(19, 19, 19)

        .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)

            .addComponent(txtBot,
javax.swing.GroupLayout.PREFERRED_SIZE,

```

```
javax.swing.GroupLayout.DEFAULT_SIZE,  
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
    .addComponent(labelBot)
```

```
    .addComponent(btnAdd)
```

```
    .addComponent(btnClear))
```

```
    .addGap(27, 27, 27)
```

```
    .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
```

```
        .addComponent(txtTop,
```

```
        javax.swing.GroupLayout.PREFERRED_SIZE,
```

```
        javax.swing.GroupLayout.DEFAULT_SIZE,
```

```
        javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
        .addComponent(labelTop)
```

```
        .addComponent(btnDelete)
```

```
        .addComponent(btnFill))
```

```
    .addGap(27, 27, 27)
```

```
    .addGroup(jPanelLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
```

```
        .addComponent(txtStep,
```

```
        javax.swing.GroupLayout.PREFERRED_SIZE,
```

```
        javax.swing.GroupLayout.DEFAULT_SIZE,
```

```
        javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
        .addComponent(labelStep)
```

```
.addComponent(btnPerform))
```

```
.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED, 49,  
Short.MAX_VALUE)
```

```
.addComponent(jScrollPane1,  
javax.swing.GroupLayout.PREFERRED_SIZE, 220,  
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
.addContainerGap())
```

```
);
```

```
javax.swing.GroupLayout layout = new  
javax.swing.GroupLayout(getContentPane());
```

```
getContentPane().setLayout(layout);
```

```
layout.setHorizontalGroup(
```

```
layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
.addComponent(jPanel, javax.swing.GroupLayout.DEFAULT_SIZE,  
javax.swing.GroupLayout.DEFAULT_SIZE,  
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
);
```

```
layout.setVerticalGroup(
```

```
layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```



```

        ));

    } else {

        JOptionPane.showMessageDialog(null, "Table is clear, you can't "

            + "entries until the button \'Fill\' has been clicked");

    }


    } catch (InvalidValueException e){

        JOptionPane.showMessageDialog(null, e.getMessage() + e.getNumber(),

"ERROR!!!", JOptionPane.ERROR_MESSAGE);

        model.removeRow(model.getRowCount() - 1);


    } catch (NumberFormatException e) {

        JOptionPane.showMessageDialog(null, "Invalid Value", "ERROR!!!",

JOptionPane.ERROR_MESSAGE);

        model.removeRow(model.getRowCount() - 1);


    } finally {

        txtBot.setText("");

        txtTop.setText("");

        txtStep.setText("");

    }

```

```
}//GEN-LAST:event_btnAddActionPerformed
```

```
private void btnDeleteActionPerformed(java.awt.event.ActionEvent evt)
```

```
{//GEN-FIRST:event_btnDeleteActionPerformed
```

```
    // TODO add your handling code here:
```

```
    DefaultTableModel model = (DefaultTableModel)jTable.getModel();
```

```
    try {
```

```
        int selectedRowID = jTable.getSelectedRow();
```

```
        model.removeRow(selectedRowID);
```

```
        functionList.remove(selectedRowID);
```

```
    } catch (Exception e){
```

```
        JOptionPane.showMessageDialog(null, e);
```

```
    }
```

```
}//GEN-LAST:event_btnDeleteActionPerformed
```

```
private void btnPerformActionPerformed(java.awt.event.ActionEvent evt)
```

```
{//GEN-FIRST:event_btnPerformActionPerformed
```

```
    // TODO add your handling code here:
```

```
    DefaultTableModel model = (DefaultTableModel)jTable.getModel();
```

```
    double result;
```

```
    int selectedRowID = jTable.getSelectedRow();
```

```

        RecIntegral integral = functionList.get(selectedRowID);

        functionList.set(selectedRowID,
integral).setBotLimit(Double.parseDouble(jTable.getValueAt(selectedRowID,
0).toString()));

        functionList.set(selectedRowID,
integral).setTopLimit(Double.parseDouble(jTable.getValueAt(selectedRowID,
1).toString()));

        functionList.set(selectedRowID,
integral).setStep(Double.parseDouble(jTable.getValueAt(selectedRowID,
2).toString()));

```

```

        result = integral.Integral();

        model.setValueAt(result, selectedRowID, 3);

        functionList.set(selectedRowID,
integral).setResult(Double.parseDouble(jTable.getValueAt(selectedRowID,
3).toString()));

```

```

} //GEN-LAST:event_btnPerformActionPerformed

```

```

private void btnClearActionPerformed(java.awt.event.ActionEvent evt) { //GEN-FIRST:event_btnClearActionPerformed

```

```

    // TODO add your handling code here:

```

```

    DefaultTableModel model = (DefaultTableModel)jTable.getModel();

```

```

        model.setRowCount(0);

        isClear = false;

    }//GEN-LAST:event_btnClearActionPerformed


    private void btnFillActionPerformed(java.awt.event.ActionEvent evt) { //GEN-FIRST:event_btnFillActionPerformed

        // TODO add your handling code here:

        DefaultTableModel model = (DefaultTableModel)jTable.getModel();

        Object[] data = new Object[4];

        for (int i = 0; i < functionList.size(); i++){

            data[0] = functionList.get(i).getBotLimit();

            data[1] = functionList.get(i).getTopLimit();

            data[2] = functionList.get(i).getStep();

            data[3] = functionList.get(i).getResult();

            model.addRow(data);

        }

        isClear = true;

    }//GEN-LAST:event_btnFillActionPerformed


    /**
     * @param args the command line arguments
     */

    public static void main(String args[]) {

```

```

/* Set the Nimbus look and feel */

//<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">

/* If Nimbus (introduced in Java SE 6) is not available, stay with the default
look and feel.

    * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html

*/

try {

    for (javax.swing.UIManager.LookAndFeelInfo info :
        javax.swing.UIManager.getInstalledLookAndFeels()) {

        if ("Nimbus".equals(info.getName())) {

            javax.swing.UIManager.setLookAndFeel(info.getClassName());

            break;

        }

    }

} catch (ClassNotFoundException ex) {

    java.util.logging.Logger.getLogger(ThirdTask.class.getName()).log(java.util.loggi
ng.Level.SEVERE, null, ex);

} catch (InstantiationException ex) {

    java.util.logging.Logger.getLogger(ThirdTask.class.getName()).log(java.util.loggi
ng.Level.SEVERE, null, ex);

} catch (IllegalAccessException ex) {

```

```
java.util.logging.Logger.getLogger(ThirdTask.class.getName()).log(java.util.loggi  
ng.Level.SEVERE, null, ex);
```

```
    } catch (javax.swing.UnsupportedLookAndFeelException ex) {
```

```
java.util.logging.Logger.getLogger(ThirdTask.class.getName()).log(java.util.loggi  
ng.Level.SEVERE, null, ex);
```

```
    }
```

```
//</editor-fold>
```

```
//</editor-fold>
```

```
/* Create and display the form */
```

```
java.awt.EventQueue.invokeLater(new Runnable() {
```

```
    public void run() {
```

```
        new ThirdTask().setVisible(true);
```

```
    }
```

```
});
```

```
}
```

```
// Variables declaration - do not modify//GEN-BEGIN:variables
```

```
private javax.swing.JButton btnAdd;
```

```
private javax.swing.JButton btnClear;
```

```
private javax.swing.JButton btnDelete;
```

```
private javax.swing.JButton btnFill;

private javax.swing.JButton btnPerform;

private javax.swing.JPanel jPanel;

private javax.swing.JScrollPane jScrollPane1;

private javax.swing.JTable jTable;

private javax.swing.JLabel labelBot;

private javax.swing.JLabel labelF;

private javax.swing.JLabel labelFunction;

private javax.swing.JLabel labelStep;

private javax.swing.JLabel labelTop;

private javax.swing.JTextField txtBot;

private javax.swing.JTextField txtStep;

private javax.swing.JTextField txtTop;

// End of variables declaration//GEN-END:variables

}
```

Файл RecIntegral.java

```
/*

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change this license

* Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this
template

*/
```



```
package ru.psu.examplefirst;
```

```
/**
```

```
*
```

```
* @author Asus
```

```
*/
```

```
public class RecIntegral{
```

```
    private double botLimit, topLimit, step, result;
```

```
    public RecIntegral(double botLimit, double topLimit, double step) throws  
    InvalidValueException {
```

```
        if (botLimit < 0.000001 || botLimit > 1000000)
```

```
            throw new InvalidValueException("The value of the bottom limit of the  
integral\n must be between 0.000001 and 1000000!!!\n"
```

```
                + "You entered: ", (int) botLimit);
```

```
        if (topLimit < 0.000001 || topLimit > 1000000)
```

```
            throw new InvalidValueException("The value of the top limit of the  
integral\n must be between 0.000001 and 1000000!!!\n"
```

```
                + "You entered: ", (int) topLimit);
```

```
        if (step < 0.000001 || step > 1000000)
```

```
            throw new InvalidValueException("The value of the step of the integral\n  
must be between 0.000001 and 1000000!!!\n"
```

```
+ "You entered: ", (int) step);
```

```
if (botLimit >= topLimit)
```

```
    throw new InvalidValueException("The value of the bottom limit must not  
be greater than the top limit!!!");
```

```
if (step >= (topLimit - botLimit) / step)
```

```
    throw new InvalidValueException("The value of the step limit must not be  
greater than integration limit!!!\n"
```

```
+ "You entered: ", (int) step);
```

```
this.botLimit = botLimit;
```

```
this.topLimit = topLimit;
```

```
this.step = step;
```

```
this.result = 0.0;
```

```
}
```

```
private static double Function(double x){
```

```
    return java.lang.Math.sqrt(x);
```

```
}
```

```
public double Integral(){
```

```
result = 0.0;
```

```
int n = (int)((topLimit - botLimit) / step);
```

```
double last = botLimit + n * step;
```

```
for (int i = 0; i < n - 1; i++){
```

```
    double x0 = botLimit + i * step;
```

```
    double x1 = botLimit + (i + 1) * step;
```

```
    result += (Function(x0) + Function(x1)) * step / 2;
```

```
}
```

```
    double x = (last < topLimit) ? ((Function(last) + Function(topLimit)) *  
(topLimit - last) / 2) : 0;
```

```
    result += x;
```

```
    return result;
```

```
}
```

```
public void setBotLimit(double botLimit){
```

```
    this.botLimit = botLimit;
```

```
}
```

```
public double getBotLimit(){
```

```
        return botLimit;
    }

    public void setTopLimit(double topLimit){
        this.topLimit = topLimit;
    }

    public double getTopLimit(){
        return topLimit;
    }

    public void setStep(double step){
        this.step = step;
    }

    public double getStep(){
        return step;
    }

    public void setResult(double result){
        this.result = result;
    }
}
```

```
public double getResult(){  
    return result;  
}  
  
}
```

Файл InvalidValueException.java

```
/*  
  
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to  
change this license  
  
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this  
template  
  
*/  
  
package ru.psu.examplefirst;  
  
  
//import javax.swing.JOptionPane;  
  
  
/**  
  
 *  
  
 * @author Asus  
  
 */  
  
public class InvalidValueException extends Exception {  
  
    private int number;
```

```
public int getNumber(){  
    return number;  
}
```

```
public InvalidValueException(String description, int number){  
    //JOptionPane.showMessageDialog(null, description, "ERROR!!!",  
JOptionPane.ERROR_MESSAGE);  
    super(description);  
    this.number = number;  
}
```

```
public InvalidValueException(String description){  
    super(description);  
}  
}
```